

Deep learning

Final project

PLANT DISEASE CLASSIFIER



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Project Overview

PROBLEM

PLANT DISEASES CAUSE 20-40% OF GLOBAL CROP LOSSES.
EARLY DETECTION PREVENTS DAMAGE.
GOAL: CLASSIFY 15 DISEASES FROM A LEAF PHOTO.

DATASET

PLANTVILLAGE: 20,638 IMAGES
15 CLASSES (TOMATO, POTATO, PEPPER)
SEVERE IMBALANCE:
152 → 3,208 PER CLASS

APPROACH

WEEK 2: BASELINE CNN
WEEK 3: PLANTCNN + AUGMENTATION
WEEK 4: RESNET18 + GRAD-CAM



Dataset

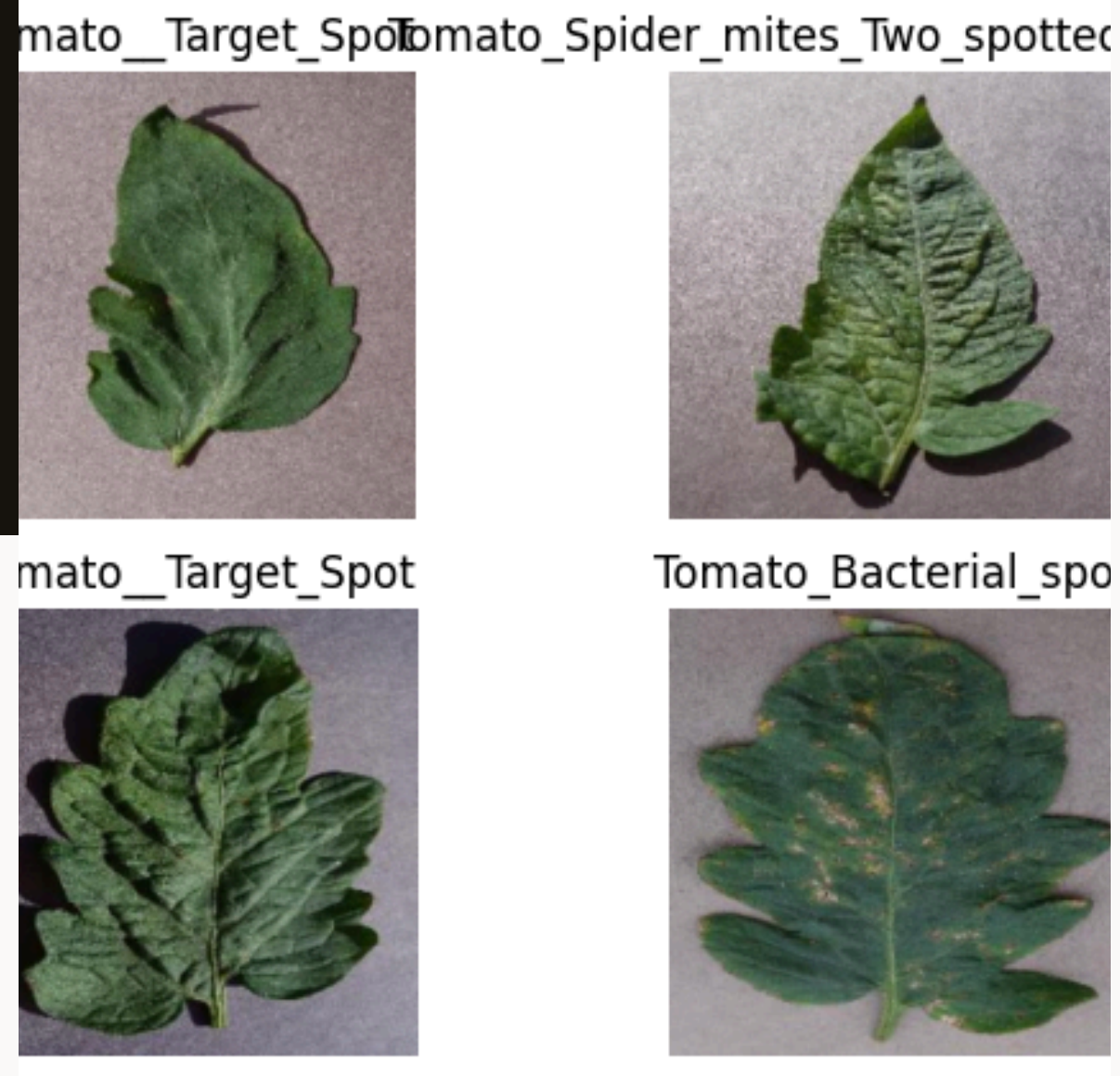
PlantVillage

20,638 - total images
15 - disease classes
3 - plant species
80/20 - train / val split



Class Imbalance
Tomato_YellowLeaf_Curl_Virus
3208
Tomato_Bacterial_spot 2127
Tomato_Spider_mites 1676
**Fixed with
WeightedRandomSampler**

Preprocessing & Augmentation



PREPROCESSING:

- Resize to 128×128 (or 224×224 for ResNet18)
- ToTensor — pixel values to [0, 1]
- Normalize: mean=[0.485, 0.456, 0.406]
- WeightedRandomSampler (class balance)
- Label smoothing $\epsilon = 0.1$

AUGMENTATION (train only)

- RandomHorizontalFlip
- RandomVerticalFlip
- RandomRotation $\pm 20^\circ$
- RandomResizedCrop



Week 2

Baseline CNN

01

INPUT $3 \times 224 \times 224$

01

CONV2D($3 \rightarrow 16$, 3×3) + RELU
+ MAXPOOL

01

CONV2D($16 \rightarrow 32$, 3×3) + RELU
+ MAXPOOL

01

FLATTEN

01

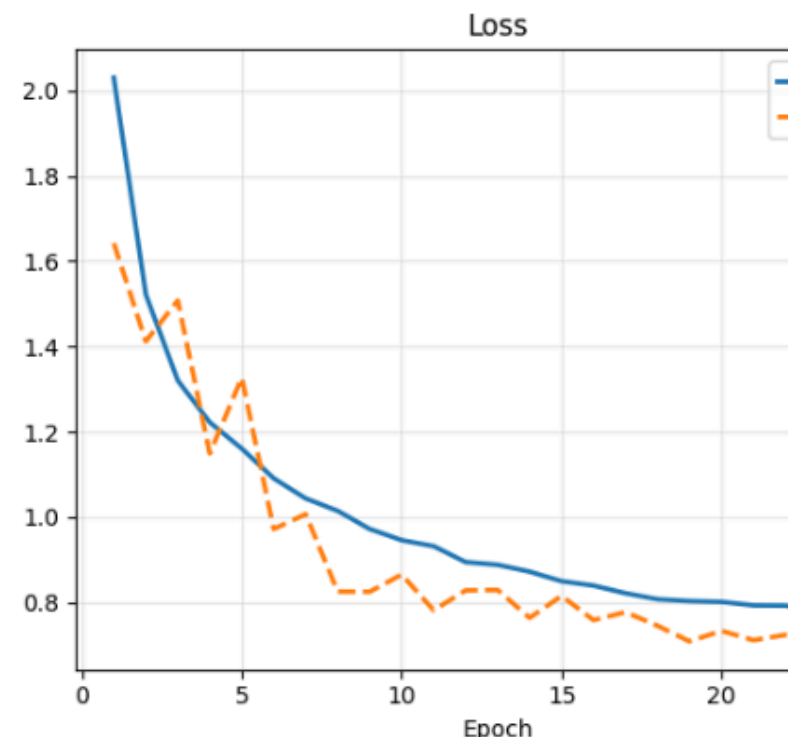
LAZYLINEAR $\rightarrow 128$
+ RELU

01

LINEAR($128 \rightarrow 15$)

Result : Error
for some reason, my
result code wasn't
working.

Week 3 PlantCNN



INPUT 3 * 128 * 128

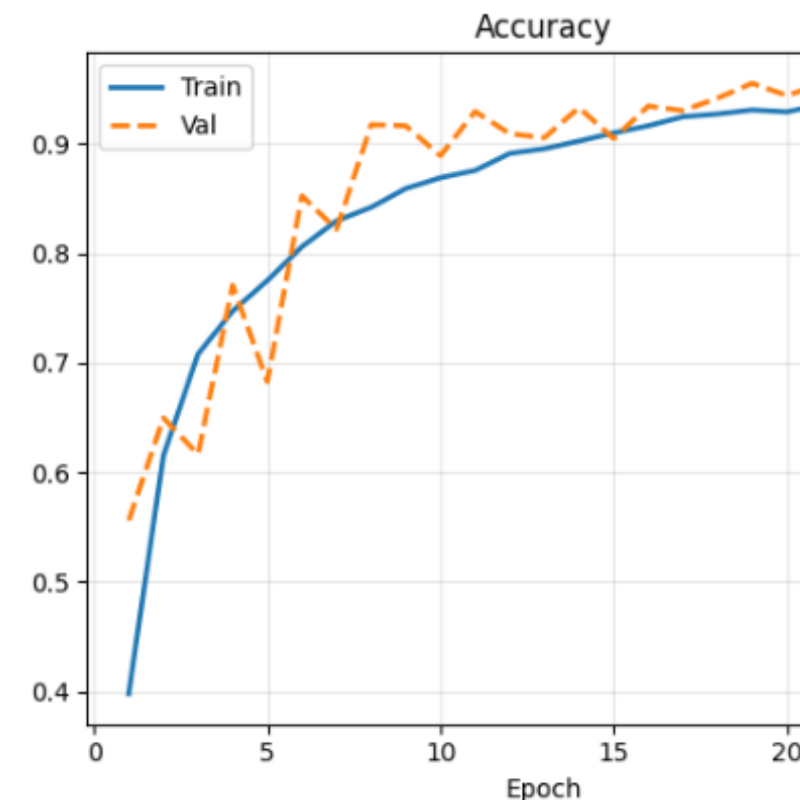
ConvBlock(3 → 32): Conv→BN→ReLU×2 + MaxPool
ConvBlock(32 → 64) + Dropout2d(0.2)
ConvBlock(64 → 128) + Dropout2d(0.2)

- Improvements vs Week 2
- BatchNorm after every Conv
 - Dropout2d + Dropout
 - Data augmentation
 - WeightedRandomSampler
 - CosineAnnealing LR

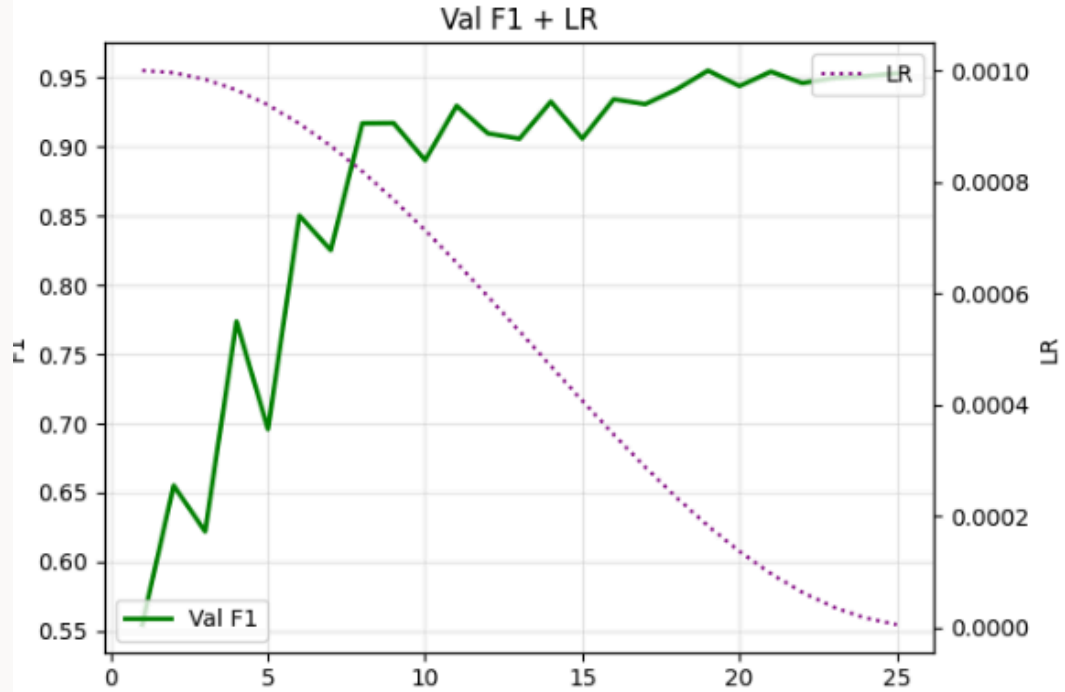
ConvBlock(128 → 256)+ Dropout2d(0.4)
GlobalAvgPool(1×1)
FC(256→512) → ReLU → Dropout(0.4)

FC(512 → 15)

0.9549 VAL ACC



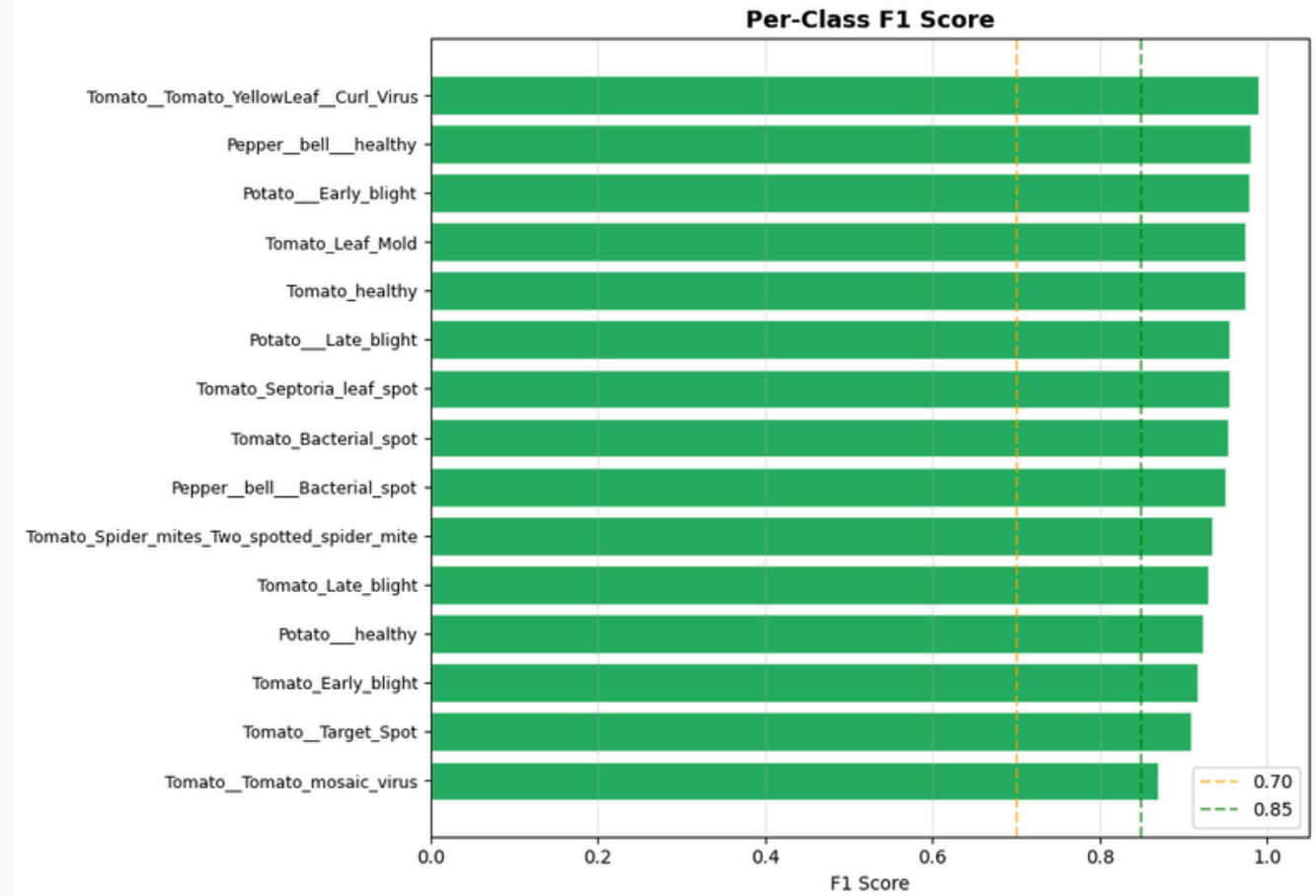
Training Results Week 3



95.49%
Val Accuracy

0.96
Weighted F1

94 min
Train Time



Best classified (F1 ≥ 0.97)	Hardest classes (F1 < 0.93)
Tomato_YellowLeaf_Curl_Virus 0.99	Tomato_mosaic_virus0.87
Pepper_bell__healthy0.98	Tomato_Target_Spot0.91

Week 4 – Transfer Learning: ResNet18

Why Transfer Learning?

ResNet18 was pre-trained on 1.2M ImageNet images. It already knows edges, textures, shapes all useful for leaf disease detection.

Stage 1 · Epochs 1–2

Backbone FROZEN | Only new head trains | LR = 1e-3

Prevents destroying ImageNet weights before head is ready

Stage 2 · Epochs 2-3

Full network UNFROZEN | Backbone LR = 1e-4, Head LR = 1e-3

Gently adapts backbone to plant disease features

ResNet18 → fc: Dropout(0.4) → Linear(512→256) → ReLU → Dropout(0.2) → Linear(256→15)

Model Comparison

	Baseline CNN (Week 2)		PlantCNN (Week 3)	ResNet18 (Week 4)
Val Accuracy	-		0.9549	0.8292
Weighted F1	-		0.96	0.81
Parameters	~200K		~1.2M	~11.2M
		Dropout	✓ (0.4)	✓ (0.4)
		Augmentation	yes	yes
		WeightedSampler	yes	yes
Training Time	~20 min		~94 min	64.2 min

Error Analysis & Limitations

Where the model fails

Tomato_mosaic_virus F1=0.87

Only 373 train images. Low precision (0.77) – over-predicts this class.

Early vs Late blight

Both brown lesion diseases. Subtle visual difference confuses the model.

Confidence ≠ correctness

Model can be 95% confident and still wrong on rare classes.

Limitations

Only 3 plants

Pepper, Potato, Tomato. Does not generalise to other crops.

Controlled images

PlantVillage has lab photos. Real farm photos may differ.

Single disease per leaf

App cannot detect multiple diseases on one image.

Conclusions

95.49% val accuracy

Weighted F1 = 0.96

Our model demonstrates promising results for real-world plant disease detection, enabling practical deployment in agricultural applications.

ResNet
Model was trained for 3 epochs only due to high training time per epoch.

Despite limited training, achieved 82.9% validation accuracy.

PlantCNN
for 3 epoch has 0.6164

ResNet will be better if i spent more times to train. but my laptop my limitations